Section 1.1 – What is Internet?

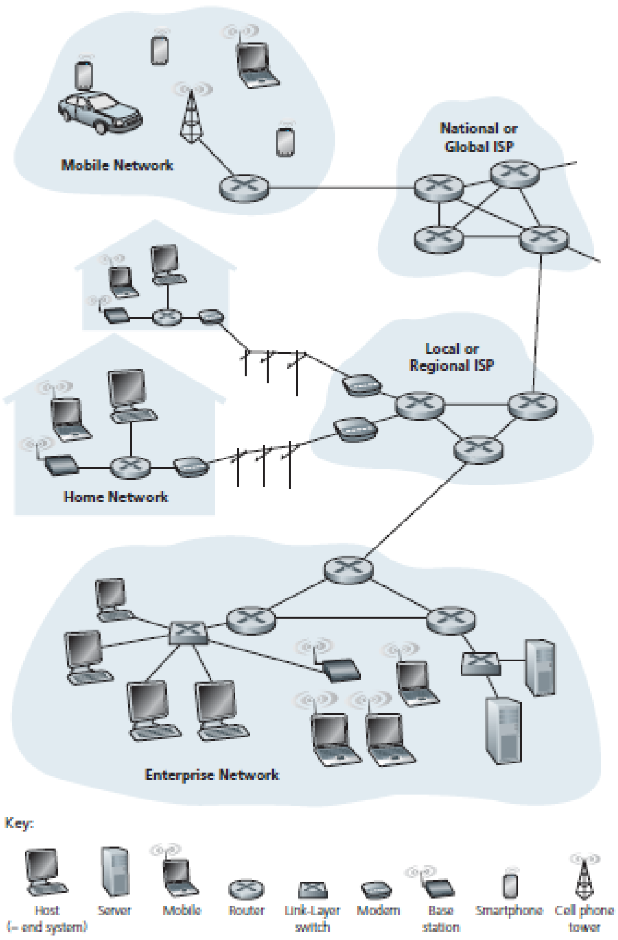


Figure 1.1 ♦ Some pieces of the Internet

- ✓ **Exercise 1:** From **figure-1.1**, Identify:
  - 4 end systems:
  - 4 intermediate devices:
  - 2 types of services by Internet:

1.1.3 What Is a Protocol?

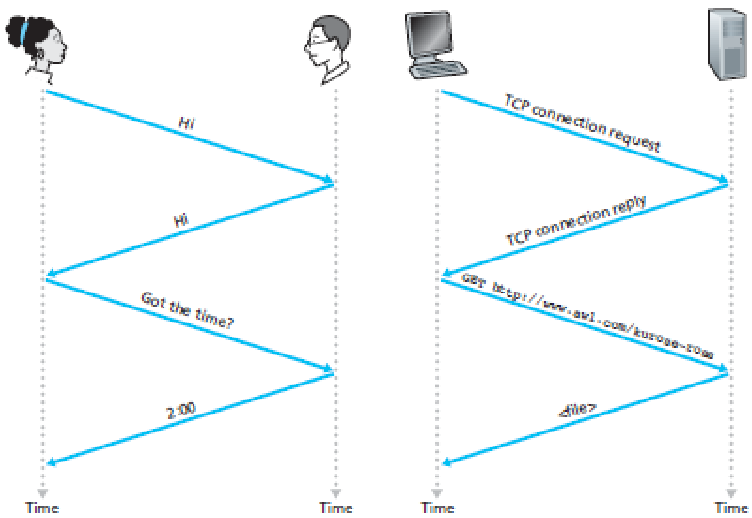


Figure 1.2 ♦ A human protocol and a computer network protocol

- ✓ **Exercise 2:**
  - What are the **3 keywords** used to define a protocol?
  - List some **well-known protocols** used in **Internet**:

Section 1.2 – The Network Edge

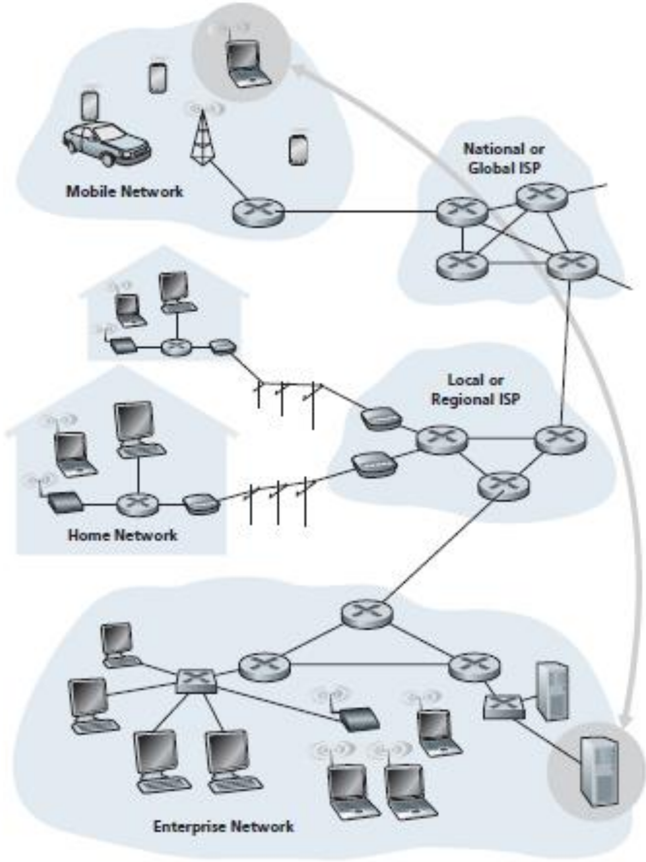


Figure 1.3 ♦ End-system interaction

- ✓ **Exercise 3:** From **figure-1.3**, Identify:
  - Network Edge:
  - Network Core:
  - 4 types of physical media:
  - 4 types of network access:

Section 1.3 – The Network Core

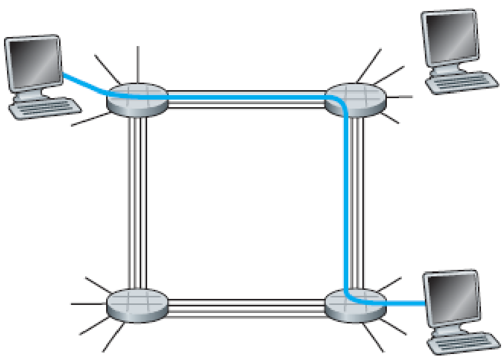


Figure 1.13 ♦ A simple circuit-switched network consisting of four switches and four links

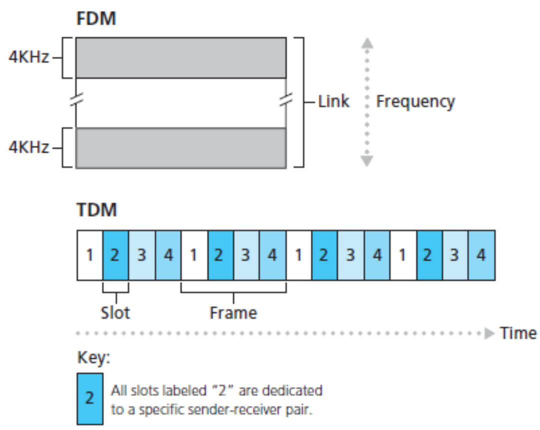


Figure 1.14 ♦ With FDM, each circuit continuously gets a fraction of the bandwidth. With TDM, each circuit gets all of the bandwidth periodically during brief intervals of time (that is, during slots)

✓ **Exercise 4:** How long does it take to send a file of **640,000 bits** from host A to host B over **a circuit-switched network**?

- ❖ All links are 1.536 Mbps
- ❖ Each link uses TDM with 24 slots/sec
- ❖ 500 msec to establish end-to-end circuit

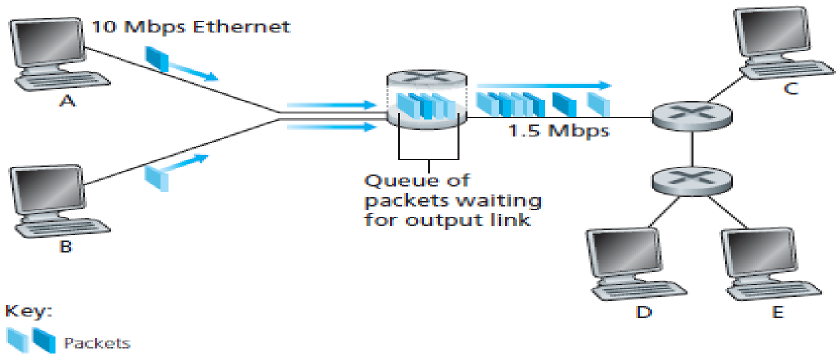
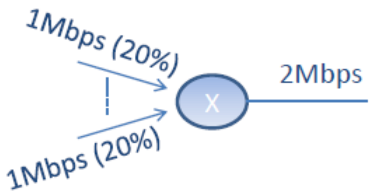


Figure 1.12 ◆ Packet switching

✓ **Exercise 5**: Suppose users share a 2Mbps (Megabits per second) link. Also suppose each user transmits continuously at 1Mbps when transmitting, but each user transmits only 20 percent of the time.

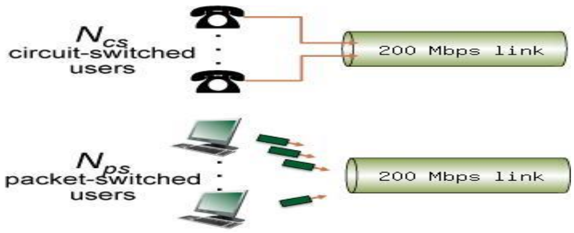
a) When **circuit switching** is used, how many users can be supported?

b) Suppose **packet switching** is used, why will there be essentially no queuing delay before the link if two or fewer users transmit at the same time?



✓ **Exercise 6**:This question requires a little bit of background in probability (but we'll try to help you though it in the solutions). Consider the **two scenarios** below:

- A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 200 Mbps.
- A packet-switching scenario with N_{ps} users sharing a 200 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 20 percent of the time.



Answer the following questions:

a. When **circuit switching** is used, what is the **maximum number of circuit-switched users** that can be supported? Explain your answer.
b. For the remainder of this problem, suppose **packet switching** is used. Suppose there are **15 packet-switching users (i.e., N_{ps} = 15)**. Can this many users be supported under circuit-switching? Explain.
c. **What is the probability** that a given (*specific*) user is transmitting, and the remaining users are not transmitting?
d. Find the probability that at any given time, **all 15 users** are transmitting simultaneously.
e. **What is the probability** that *more* than 8 users are transmitting? Comment on what this implies about the number of users supportable under circuit switching and packet switching.

Knowledge Test # 1

Q.1 What is the difference between **a host** and **an end system**? List different types of end systems. Is a **Web server** an end system?

Q.2 List **six access technologies**. Classify each one as **home access**, **enterprise access**, or **wide-area wireless** access.

Q.3 What **advantage** does a **circuit-switched network** have over a **packet-switched network**? What **advantage** does a **packet-switched network** have over a **circuit-switched network**? What advantages does **TDM** have over **FDM** in a **circuit-switched network**?

Q.4 Suppose users share a **10 Mbps** link. Also suppose each user transmits continuously at **1 Mbps** when transmitting, but each user transmits only **20 percent** of the time.

a. When **circuit switching** is used, **how many users** can be supported?

b. For the remainder of this problem, suppose **packet switching** is used and there are **20 users**. Why will there be essentially **no queuing delay** before the link if **ten or fewer** users transmit at the same time? Why will there be a queuing delay if **more than 10 users** transmit at the same time?

c. Find the probability that **a given user** is transmitting.

d. Find the probability that **a given user** is transmitting and **others are not** transmitting.

e. Find the probability that **any one user** is transmitting and **others are not** transmitting.

f. Find the probability that at any given time, all **20 users** are transmitting simultaneously.

g. Find the probability that more than **10 users** are transmitting simultaneously (**fraction of time** during which the queue grows).

Section 1.4 – Delay, Loss, and Throughput in Packet-Switched Networks

❖ **Four sources of packet delay**

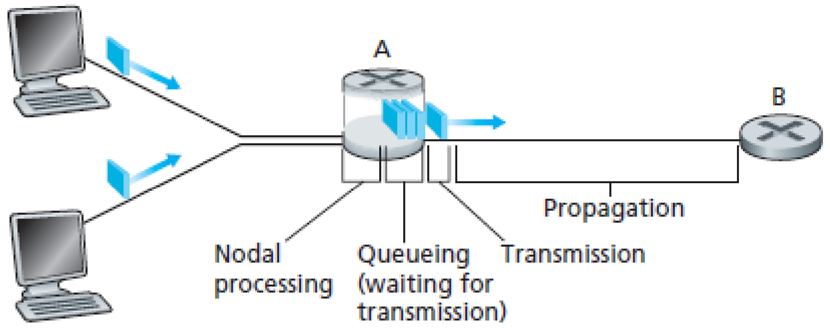


Figure 1.16 ♦ The nodal delay at router A

❖ **Comparing Transmission and Propagation Delay**

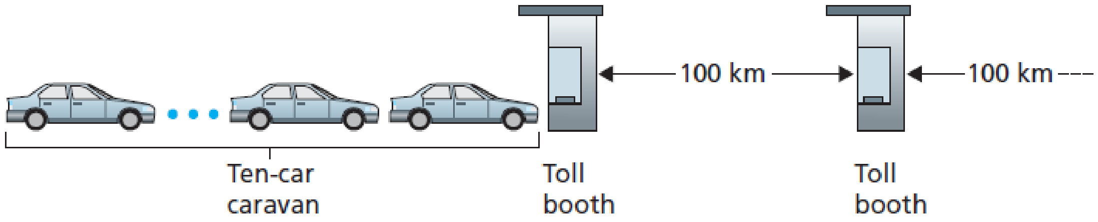
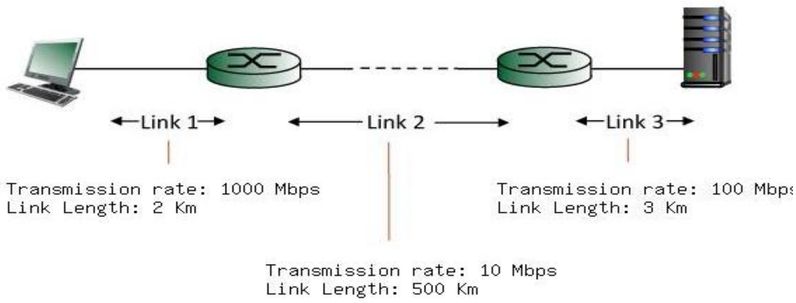


Figure 1.17 ♦ Caravan analogy

✓ **Exercise 7:** Consider the **caravan analogy** example shown in **figure 1.17**. Assume, propagation speed of a car is **200 km/hour**, caravan travels **20 km starting from 1st toll booth**, and also there are **eight cars** in the caravan. Assuming **2 minute service time** for each car in the toll booth:

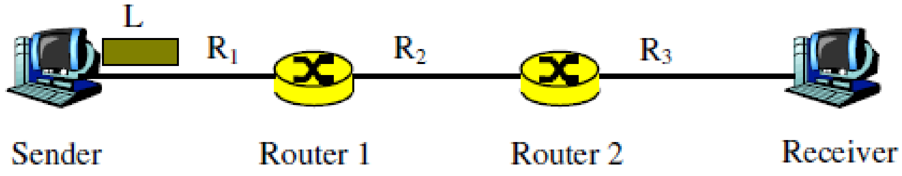
- What is the **end-to-end delay** (from 1st toll booth to 3rd toll booth)?
- **How many cars** will reach toll booth 2 before all the cars are serviced by toll booth 1?

✓ **Exercise 8:** Consider the figure below, with **three links**, each with the specified transmission rate and link length.

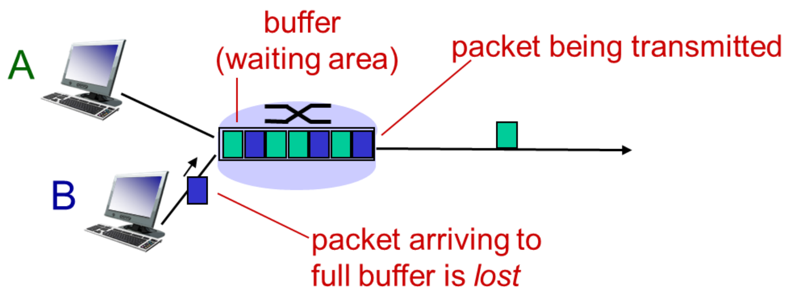


Find the **end-to-end delay** (including the transmission delays and propagation delays on each of the three links, but ignoring queuing delays and processing delays) from when the left host begins transmitting the first bit of a packet to the time when the last bit of that packet is received at the server at the right. The speed of light propagation delay on each link is 3x10**8 m/sec. Note that the transmission rates are in Mbps and the link distances are in Km. Assume a packet length of **16000** bits. Give your answer in **milliseconds**.

✓ **Exercise 9:** Consider following **figure**, where the sender transmits one packet of size ***L* = 1 KB**(kilobytes). The **link speeds** are ***R*1 = *R*2 = *R*3 = 1 Mbps** (megabits per second). The **length of each link** is**100 Km**. Compute the **end-to-end delay** for the following situations. We assume that the **propagationspeed** of the packet is **20,000 Km/s**, and that the **processing time** of the packet at each router is **5 µs**, and both routers are **store-and-forward** routers.



❖ Packet Loss



❖ Throughput

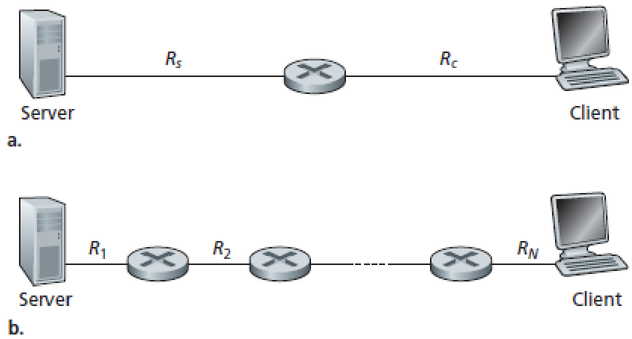


Figure 1.19 ♦ Throughput for a file transfer from server to client

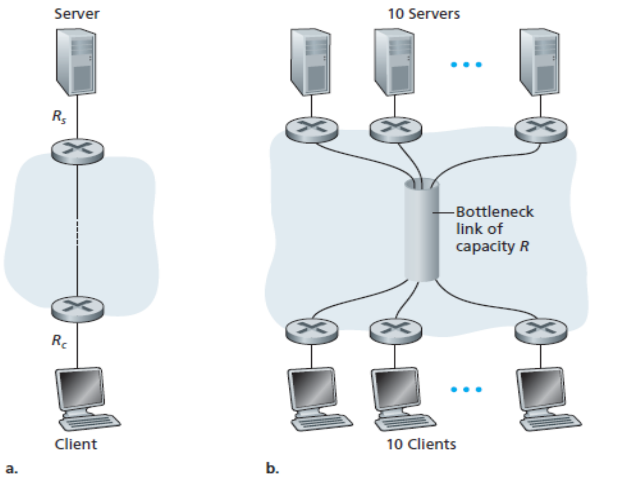


Figure 1.20 ♦ End-to-end throughput: (a) Client downloads a file from server; (b) 10 clients downloading with 10 servers

✓ **Exercise 10:** Suppose **Host A** wants to send a large file to **Host B**. The path from Host A to Host B has **three links**, of rates **R1 = 500 kbps**, **R2 = 2 Mbps**, and **R3 = 1 Mbps**.

a. Assuming **no other traffic** in the network, what is the **throughput** for the file transfer.

b. Suppose the file is **4 million bytes**. Dividing the file size by the throughput, roughly **how long** will it take to transfer the file to Host B?

c. Repeat (a) and (b) now with **R2** reduced to **100 kbps**.

Knowledge Test # 2

Q.1 What are the **3 basic parameters** used to measure **performance** of a packet-switched network?

Q.2 What are the **four sources** of packet delay in a node?

Q.3 Consider a packet of **length *L*** which begins at end system A and travels over **three links** to a destination end system. These three links are connected by **two packet switches**. Let d_i, s_i, and R_i denote the length, propagation speed, and the transmission rate of link *i,* for *i*= 1, 2, 3. Suppose now the packet is **1,500 bytes**, the **propagation speed** on all three links is **$2.5x10^8$ m/s**, the **transmission rates** of all three links are **2 Mbps**, the packet switch **processing delay** is **3 msec**, the length of the first link is **5,000 km**, the length of the second link is **4,000 km**, and the length of the last link is **1,000 km**. For these values, what is the **end-to-end delay**?

Q.4 A **packet switch** receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, **one other packet** is halfway done being transmitted on this outbound link and **200 other packets are waiting to be transmitted**. Packets are transmitted in order of arrival. Suppose all packets are **64 bytes** and the link is **10 Mbps**. What is the **queuing delay** for the packet?

Q.5 A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is **halfway done** being transmitted on this outbound link and **four other packets** are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are **1,500 bytes** and the link rate is **2 Mbps**. What is the **queuing delay** for the packet? More generally, what is the queuing delay when all packets have length *L*, the transmission rate is *R*, *x* bits of the currently being- transmitted packet have been transmitted, and *n* packets are already in the queue?

Q.6 How long does it take a packet of length **1.45 kbytes** to propagate down over a link of **distance 500 km**, propagation speed **$2.5x10^8$ m/s**, and transmission rate **10 Gbps**?

Q.7 Suppose you would like to urgently deliver **40 terabytes** (40×10^{12} bytes) data from Dhaka to Chittagong. You have available a **1 Gbps dedicated link** for data transfer. Would you prefer to transmit the data via this link or instead use **FedEx overnight delivery**? **Explain**.

Q.8 Suppose there is a **10 Mbps microwave link** between a **geostationary satellite** (Recall geostationary satellite is **36,000 kilometers** away from earth surface) and its base station on Earth. **Every minute** the satellite takes a digital photo and sends it to the base station. Assume a propagation speed of **2.4×10^8 m/s**.

1. What is the **propagation delay** of the link?
2. Let ***x*** denote the size of the photo. What is the **minimum value of *x*** for the microwave link to be continuously transmitting?

Section 1.5 – Protocol Layers and Their Service Models

- ✓ **Why Layering?**
  - ✓ **Real life examples:** Postal service, Air Travel, Office-to-office communication

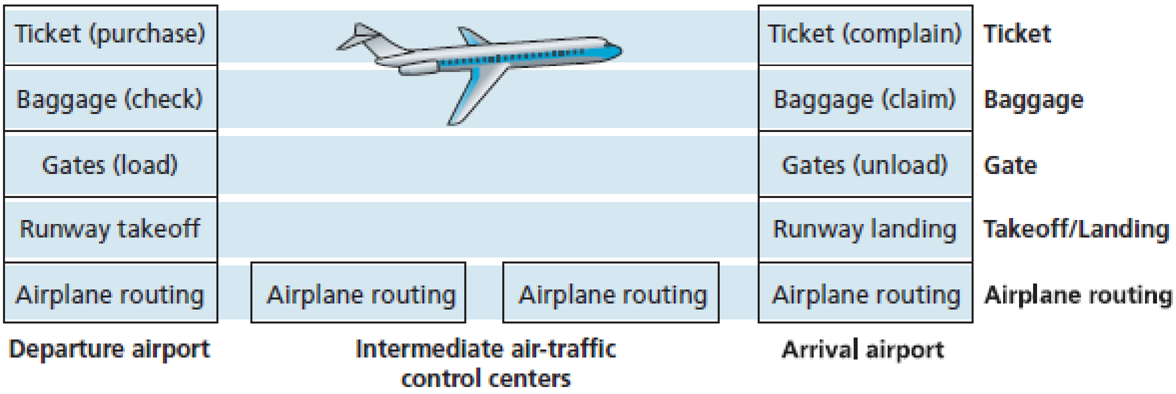
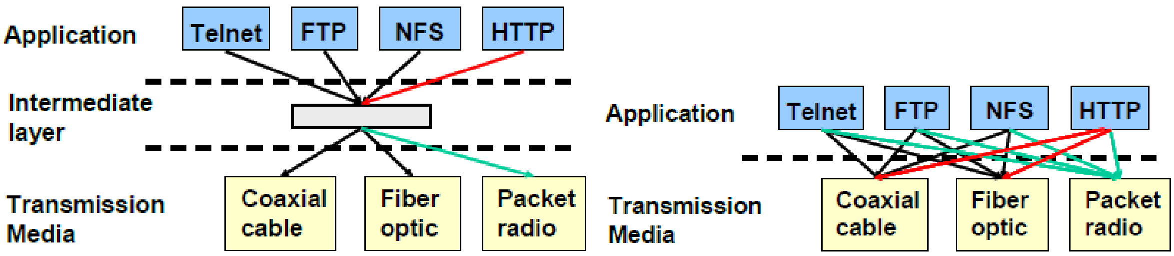


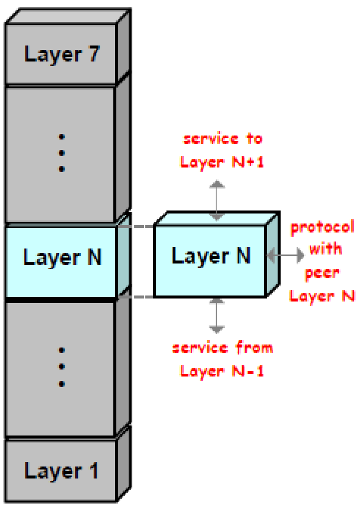
Figure 1.22 ♦ Horizontal layering of airline functionality

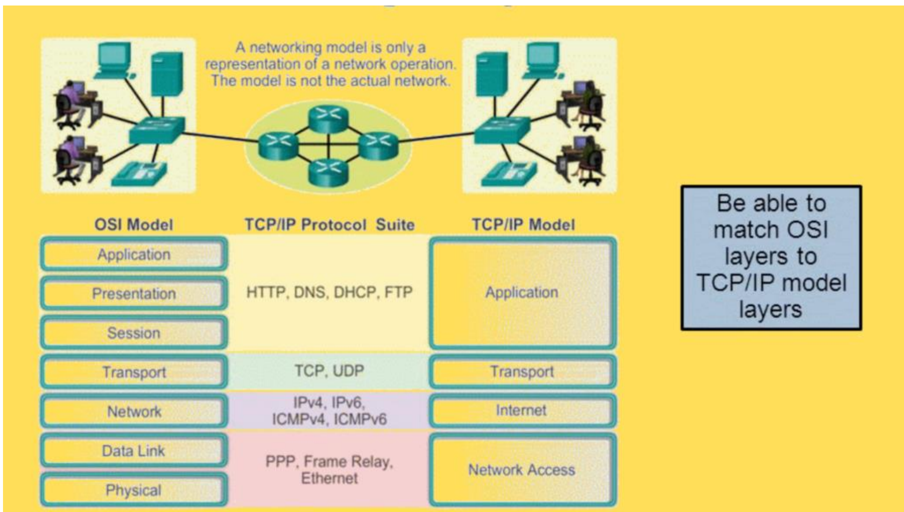
  - ✓ **Computer network example:** Development of a new Web Browser (Layering vs. No layering)



  - ✓ **Advantages of layering architecture:** Modularity, trouble shooting, functionality reuse
  - ✓ **Monolithic network design** (1 large main function) **vs. Layered approach** (Main function calling smaller functions)

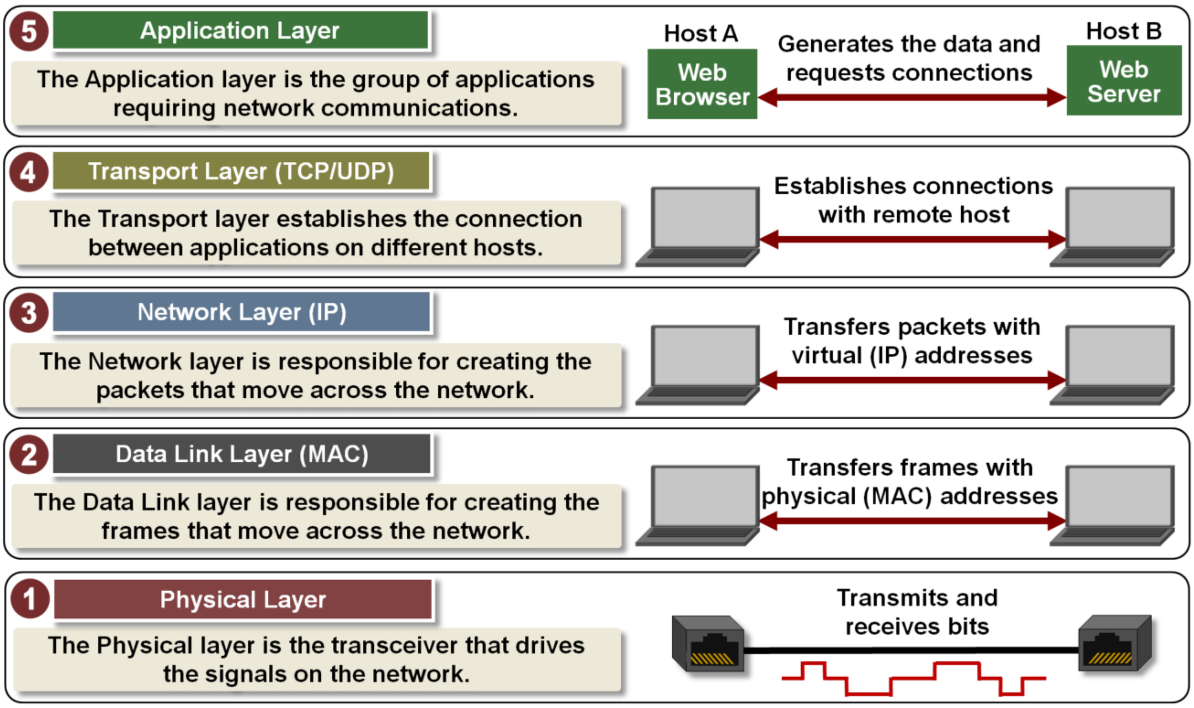
- ✓ **Vertical vs. Horizontal communication**



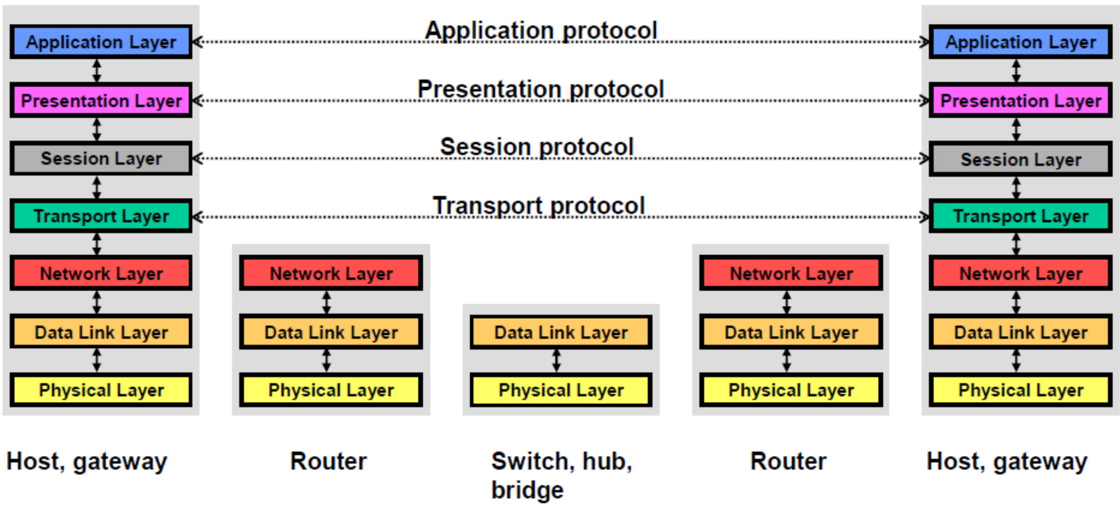


✓ OSI Layering Model:

L #	Layer Name	Basic functions	PDU	Encapsulation
7	Application	Network services (email, file transfer)	Data	AH+Data
6	Presentation	Formatting, encryption, and compression	Data	PH+AH+data
5	Session	Setup and management of end-to-end conversation	Data	SH+PH+AH+data
4	Transport	End-to-end delivery of messages	Segmant	TH+ SH+PH+AH+data
3	Network	End-to-end transmission of packets	Packet	NH+TH+ SH+PH+AH+data
2	Data link	Transmission of packets on one given link	Frame	DH+NH+TH+ SH+PH+AH+data+DT
1	Physical	Transmission of bits	Signal	Bit streams (01100101011)



✓ **Example topology:** Communication using **OSI model** (See animation: **OSI_layering_model.swf**)

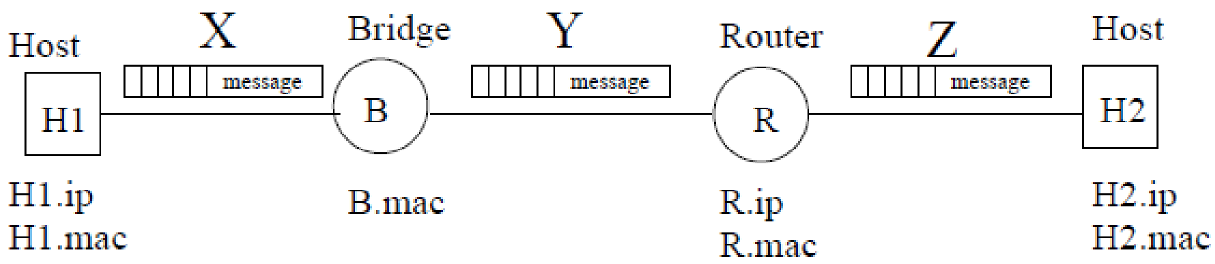


- ✓ **Exercise 11:** To reduce the complexity of the system, the **Internet** is based on a set of layered protocols and the payload is encapsulated/decapsulated whenever it is transferred from one layer to another. Think of **another real-world system** that is structured in layers and uses encapsulation/decapsulation whenever the payload is transferred from one layer to another. **Describe the layers** of the system and the encapsulation-decapsulation process. (Hint: a **postal service**)

- ✓ **Exercise 12: Understanding Packet Headers**

  Consider a message being transmitted along the path as shown in the following figure. The message was sent from an application running on Host 1 (H1) port 8888 to an application running on Host 2 (H2) port 9999. The transport protocol being used is TCP. The figure shows one packet of the message at several points along its path (points X, Y and Z). The questions will ask you what is stored in the headers at these three points.

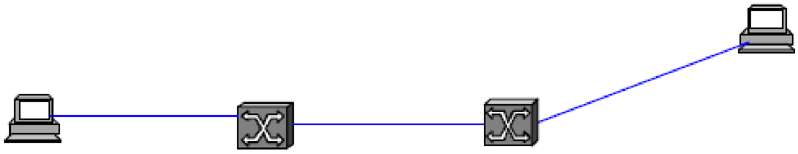
  Start by numbering the headers in the packet at points X, Y and Z. Each header should be assigned a number from 1 to 5: 1 corresponds to Physical layer, 2 corresponds to Data Link layer, 3 corresponds to Network layer, 4 corresponds to Transport layer and 5 corresponds to Application layer. For all the questions below, your answer should take the form of a list within { }. For example, (Packet X, Layer 3), (Packet Y, Layer 2)}. It can be null { } too.



  a) List all the places where **H2.ip** is written.
  b) List all the places where **R1.ip** is written.
  c) List all the places where **B.mac** is written.
  d) List all the places where **R1.mac** is written.
  e) List all the places where **H2.mac** is written.
  f) List all the places where port **8888** is written.
  g) List all the places where port **9999** is written.

- ✓ **Exercise 13:** Suppose a process wants to send an ***L*-byte message** to its peer process, using an existing TCP connection. The **TCP segment** consists of the message plus **20 bytes of header**. The segment is encapsulated into an **IP packet** that has an additional **20 bytes of header**. The IP packet in turn goes inside an **Data Link frame** that has **18 bytes of header** and **trailer**. What **percentage of the transmitted bits** in the physical layer corresponds to the message information if L = 500 bytes?

✓ **Exercise 14:** Consider a message of **150,000 bytes** that must be sent from a source A to a destination B, as a succession of PDUs made of a header containing **200 bytes** of Protocol Control Information and a fraction of the message. Each PDU is received, stored and forwarded by **2 packet switching nodes**, in order to reach its destination.



Consider the **3 scenarios**:

i. each PDU contains **10,000 bytes** of the original **150,000 bytes message**
ii. each PDU contains **2500 bytes** of the original **150,000 bytes message**
iii. each PDU contains **750 bytes** of the original **150,000 bytes message**

Which scenario leads to the shortest latency for the complete reception of the message (How long does it take in bytes to receive the complete message)? Why is it not necessarily the shortest PDU that gives the smallest latency?

✓ **Exercise 15:** Which of the following are benefits of using a **layered network model**:

i. It speci_es how changes in one layer propagate to changes in another layer.
ii. It facilitates troubleshooting
iii. It focuses on details rather than general functions of the network
iv. It breaks networking down into manageable modules
v. It allows layers developed by different vendors to interoperate

Knowledge Test # 3

Q.1 How using layering architecture simplifies network operations? Specify **3 distinct advantages** and **justify** with examples.

Q.2 List **five tasks** that a layer can perform. Is it possible that **one (or more) of these tasks** could be performed by **two (or more) layers**? If yes, give proper example.

Q.3 What is an **application-layer message**? A **transport-layer segment**? A **network layer datagram**? A **link-layer frame**?

Q.4 Which layers in the Internet protocol stack does **a router** process? Which layers does **a link-layer switch** process? Which layers does **a host** process?

Q.5 Find corresponding **layer(s)** of **OSI model** for every function in the following list:

i. Flow control.
ii. Encryption and compression of application data.
iii. Ensure reliable transmission of data between processes in different machines.
iv. Establishes, manages and terminates sessions.
v. Defines frames.
vi. Carry signals of raw data.
vii. Provides reliable end-to-end communication.
viii. Represent bytes as different voltages.
ix. Provides reliable transfer of data between two adjacent nodes.
x. Decides the path a packet will take across the network.
xi. Which layer is responsible for converting data packets from the Data Link layer into electrical signals?
xii. Which layer defines how data is formatted, presented, encoded, and converted for use on the network?
xiii. Which layer ensures the reliable transmission of data across a link and is primarily concerned with physical addressing, network topology, error detection, ordered delivery of frames, and flow control?
xiv. Which layer is used for reliable communication between end users over the network and controlling the end to end flow of information?
xv. Which layer specifies voltage, wire bandwidth, and cables and moves bits between devices?
xvi. Which layer is responsible for keeping the data from different applications separate on the network?

Q.6 List **similarities** and **differences** between **OSI** and **TCP/IP** reference models.

Q.7 Fill up the following table with **contents specified in each column**:

OSI Layer	TCP/IP layer	Why used (Main functions)	PDU Name	Encapsulation / Header	Example protocols

Q.8 Use the above filled table to describe the flow of data from **Host A** to **Host B** in the following topology:

HOST A → SWITCH 1 → ROUTER → SWITCH 2 → HOST B